

LA HW

Included exercises: 1, 3, 5, 9, 11, 15, 17, 19, 21, 23, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 49, 51

Exercises 1–8: Testing an Inverse

For each pair of matrices A and B , determine whether $B = A^{-1}$.

Original Exercise 1

$$A = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0.5 \\ 1 & -1 \end{bmatrix}.$$

Original Exercise 3

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 2 & 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 5 & -3 \\ 0 & -2 & 1 \\ -1 & -1 & 1 \end{bmatrix}.$$

Original Exercise 5

$$A = \begin{bmatrix} 1 & -2 & 1 & 0 \\ 2 & -2 & 1 & 0 \\ 1 & -1 & 0 & -1 \\ -1 & 0 & -1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 \\ 2 & -3 & 2 & -2 \\ -1 & 2 & -2 & 1 \end{bmatrix}.$$

Exercises 9–14: Expressions Involving Inverses

Find each matrix expression, where A and B are invertible 3×3 matrices satisfying

$$A^{-1} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 1 \\ 1 & 1 & -1 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 0 & 4 \\ 3 & -2 & 1 \end{bmatrix}.$$

Original Exercise 9 Find $(A^T)^{-1}$.

Original Exercise 11 Find $(AB)^{-1}$.

Exercises 15–22: Elementary Matrices

Find the inverse of each elementary matrix.

Original Exercise 15

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

Original Exercise 17

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Original Exercise 19

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Original Exercise 21

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

Exercises 23–32: Elementary Row Operations

Find an elementary matrix E such that $EA = B$.

Original Exercise 23

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & -2 \\ 3 & 4 \end{bmatrix}.$$

Original Exercise 27

$$A = \begin{bmatrix} 3 & 2 & -1 \\ -1 & 0 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 0 & 6 \\ 3 & 2 & -1 \end{bmatrix}.$$

Original Exercise 29

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & -3 & 1 \\ 0 & 4 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 & 2 \\ 2 & -3 & 1 \\ -10 & 19 & 0 \end{bmatrix}.$$

Original Exercise 31

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 7 & 8 & 9 \\ 4 & 5 & 6 \end{bmatrix}.$$

Exercises 33–52: True or False

Determine whether each statement is true or false.

Original Exercise 33 Every square matrix is invertible.

Original Exercise 35 Elementary matrices are invertible.

Original Exercise 37 If B and C are inverses of a matrix A , then $B = C$.

Original Exercise 39 An invertible matrix may have more than one inverse.

Original Exercise 41 For any matrices A and B , if A is the inverse of B^T , then A is the transpose of B^{-1} .

Original Exercise 43 If A and B are invertible $n \times n$ matrices, then $(AB)^{-1} = A^{-1}B^{-1}$.

Original Exercise 45 An elementary $n \times n$ matrix has at most $n + 1$ nonzero entries.

Original Exercise 49 If R is the reduced row echelon form of A , then there exists an invertible matrix P such that $PA = R$.

Original Exercise 51 The pivot columns of a matrix are linearly dependent.